

Sr.	Unit	Questions	Marks	S	CO																
1	1	Define DOS. What are the advantages of distributed system over centralized system?	3	R	CO 1																
2	1	Give architecture of Distributed Operating System	7	R	CO 1																
3	1	Give the classification of DOS based on instruction and data stream.	7	U	CO 1																
4	1	What are the distributed computing models? Explain workstation-server model and processor pool model in detail.	7	U	CO 1																
5	1	What are the differences between network operating system and distributed operating system?	4	U	CO 1																
6	1	Explain Design Issues/challenges of Distributed operating system	7	R	CO 1																
7	2	Explain synchronous communication and asynchronous communication.	3	U	CO 2																
8	2	What are the message buffering strategies? Explain any two.	4	R	CO 2																
9	2	How implicit addressing and explicit addressing techniques work?	7	U	CO 2																
10	2	Describe IPC protocol.	4	U	CO 2																
11	2	Explain Remote procedure call (RPC) model.	7	U	CO 2																
12	2	Discuss stateful file server and stateless file server	3	R	CO 2																
13	2	List the differences between RMI and RPC.	4	U	CO 2																
14	2	Explain Remote method invocation (RMI) implementation.	7	U	CO 2																
15	3	Define: Clock synchronization, clock skew, Drifting of clock, mutual exclusion and critical section	1	R	CO 3																
16	3	Explain Berkeley Algorithm for physical clock synchronization	7	U	CO 3																
17	3	Explain Lamport's Algorithm for logical clock synchronization	7	U	CO 3																
18	3	How Centralized Mutual Exclusion Algorithm works in distributed system.	4	U	CO 3																
19	3	Determine how token ring algorithm works for mutual exclusion?	4	U	CO 3																
20	3	Describe Bully's election algorithm by illustrating it for 7 processes group.	7	U	CO 3																
21	3	Write about name server, name agent and name resolution in naming system.	1	R	CO 3																
22	3	p1, p2 and p3 are processes of distributed system, Resource requirement of each process is shown in table. Using deadlock avoidance algorithm, check that sequence p1 p2 p3 and sequence p3 p1 p2 are safe or not? Total resources = 10	4	A	CO 3																
<table><tr><td></td><td>p1</td><td>p2</td><td>p3</td></tr><tr><td>Max resources</td><td>5</td><td>4</td><td>9</td></tr><tr><td>Holds resources</td><td>3</td><td>2</td><td>3</td></tr><tr><td>Free resources</td><td colspan="3">2</td></tr></table>				p1	p2	p3	Max resources	5	4	9	Holds resources	3	2	3	Free resources	2					
	p1	p2	p3																		
Max resources	5	4	9																		
Holds resources	3	2	3																		
Free resources	2																				
23	3	Determine how system oriented names are useful in distributed system?	4	R	CO 3																
24	4	Describe the architecture of Distributed shared memory	7	R	CO 4																
25	4	Differentiate message passing and shared memory concept.	3	U	CO 4																
26	4	What are the types of distributed shared memory, explain in detail	7	R	CO 4																
27	4	Explain consistency models in detail	7	U	CO 4																
28	4	Describe load balancing approach	4	R	CO 4																

Subject : 2301CS718 – Distributed Operating System

Prof. Nidhi K. Chitroda

29	4	What are the load sharing policies?	4	R	CO 4
30	4	What are the fault tolerance strategies?	3	R	CO 4
31	4	Define Thrashing and Process migration.	4	R	CO 4
32	5	What are the types of attacks? Explain in detail	7	U	CO 5
33	5	Differentiate symmetric and asymmetric crypto system.	7	U	CO 5
34	5	Define: Cryptography, Authentication, Authorization, Integrity, Confidentiality	1 (each)	R	CO 5
35	5	Describe secure group communication	4	U	CO 5
36	5	Write short note on firewalls	3	R	CO 5
37	5	How does Diffie-Hellman Algorithm for Key Establishment work?	7	R	CO 5